Sr. Backend Engineer 2nd Stage Interview Assignment

Context

At Formunauts, among other things, we process and analyze large volumes of tracking events generated by fundraising activities. Offerings like our Territory Manager rely on **efficient querying of geospatial and temporal data**.

This assignment simulates a simplified version of such a use case.

Objective

Implement an API endpoint that returns events occurring **near a given base event**, optionally filtered by date.

Provided Data

You are given a JSON fixture containing tracking events (see at the very end)

- Each event includes:
  - metadata (customer, fundraiser, etc.)
  - timestamp
  - geolocation (`device_coordinates` in `SRID=4326;POINT (lng lat)` format)

Use this dataset as your initial data source.

Task

Implement an API endpoint:

```
GET /events/{event_id}/nearby_events/?radius=<meters>&date=<YYYY-MM-DD>
```

Functional Requirements

- Use the event identified by `{event_id}` as the **base event**
- Return all other events:
  - within the given `radius` (in meters)
  - excluding the base event itself
- Optional:
  - filter results by `date` (UTC, based on event timestamp)

Response Format

Return results as JSON in the following structure:

```
{
  "pk": 9801211,
  "customer": 9,
  "fundraiser": 37096,
  "name": "location-check-in",
  "event_type": "automatic",
  "timestamp": "2026-03-01T14:45:12.332Z",
  "event_context": { ... },
  "browser_location": { ... },
  "device_coordinates": "SRID=4326;POINT (16.350123 48.209845)"
}
```

Technical Expectations

You are free to choose your tech stack. However, we are explicitly interested in:

1. Geospatial handling

2. API design

3. Code quality

4. Testing

5. Documentation

Design Considerations (Important)

In addition to the implementation, include **scaling & production considerations**

Please briefly describe

- How your solution would perform with **millions of events**
- How you would
  - optimize geospatial queries
  - handle indexing
  - ensure performance (e.g. <200ms response time)
- What you would change in a production system

Time Expectation

We expect this task to take approximately **2–3 hours**.

We are **not** looking for a fully production-ready system, but

- clarity of thought
- reasonable trade-offs
- ability to communicate decisions

Bonus (Optional)

- Pagination support
- API documentation (e.g. OpenAPI/Swagger)
- Use of geospatial indexing
- Containerization (Docker)

Questions

If anything is unclear, feel free to ask.

JSON Fixture

```
[
{
```

```
  "model": "tracking.event",
  "pk": 9801193,
  "fields": {
    "customer": 9,
    "fundraiser": 146,
    "name": "login",
    "event_type": "automatic",
    "timestamp": "2026-03-01T09:12:44.201Z",
    "event_context": {
      "source_id": "login",
      "target_id": null,
      "source_category": "login"
    },
    "browser_location": {
      "current": "https://staging.formunauts.com/auth/post-login",
      "referrer": "https://staging.formunauts.com/auth/login"
    },
    "device_coordinates": "SRID=4326;POINT (16.374211 48.225678)"
  }
},
{
  "model": "tracking.event",
  "pk": 9801211,
  "fields": {
    "customer": 9,
    "fundraiser": 37096,
    "name": "location-check-in",
    "event_type": "automatic",
    "timestamp": "2026-03-01T14:45:12.332Z",
    "event_context": {
      "source_id": "location-check-in",
      "target_id": null,
      "source_category": "location-check-in"
    },
    "browser_location": {
      "current": "https://staging.formunauts.com/next/(locations/21914//check-in:603/locations)",
      "referrer": "https://staging.formunauts.com/next/(locations//check-in:603/locations)"
    },
    "device_coordinates": "SRID=4326;POINT (16.350123 48.209845)"
  }
},
{
  "model": "tracking.event",
  "pk": 9801212,
  "fields": {
    "customer": 9,
    "fundraiser": 37096,
    "name": "login",
    "event_type": "automatic",
    "timestamp": "2026-03-02T10:05:59.011Z",
    "event_context": {
      "source_id": "login",
      "target_id": null,
      "source_category": "login"
    },
    "browser_location": {
      "current": "https://staging.formunauts.com/auth/post-login",
      "referrer": "https://staging.formunauts.com/auth/login"
    },
    "device_coordinates": "SRID=4326;POINT (16.368776 48.217342)"
  }
},
{
  "model": "tracking.event",
  "pk": 9801213,
  "fields": {
    "customer": 9,
    "fundraiser": 37096,
    "name": "conversation-start",
    "event_type": "click",
    "timestamp": "2026-03-02T16:22:18.445Z",
    "event_context": {
      "tag_name": "BUTTON",
      "source_id": "603",
      "target_id": null,
      "source_category": "campaign-start"
    },
```

```
    "browser_location": {
      "current": "https://staging.formunauts.com/next/forms/603/start",
      "referrer": "https://staging.formunauts.com/next"
    },
    "device_coordinates": "SRID=4326;POINT (16.355678 48.202611)"
  }
},
{
  "model": "tracking.event",
  "pk": 9801214,
  "fields": {
    "customer": 9,
    "fundraiser": 37096,
    "name": "campaign-form-open",
    "event_type": "click",
    "timestamp": "2026-03-03T08:11:05.987Z",
    "event_context": {
      "tag_name": "BUTTON",
      "source_id": "603",
      "target_id": null,
      "source_category": "campaign-start"
    },
    "browser_location": {
      "current": "https://staging.formunauts.com/next/forms/603/start",
      "referrer": "https://staging.formunauts.com/next"
    },
    "device_coordinates": "SRID=4326;POINT (16.361122 48.222109)"
  }
},
{
  "model": "tracking.event",
  "pk": 9801215,
  "fields": {
    "customer": 9,
    "fundraiser": 37096,
    "name": "campaign-form-success",
    "event_type": "click",
    "timestamp": "2026-03-03T11:54:33.210Z",
    "event_context": {
      "tag_name": null,
      "source_id": "603",
      "target_id": null,
      "source_category": "campaign-form"
    },
    "browser_location": {
      "current": "https://staging.formunauts.com/next/forms/603/form/pages/signature-step",
      "referrer": "https://staging.formunauts.com/next/forms/603/form/pages/qa-step"
    },
    "device_coordinates": "SRID=4326;POINT (16.378901 48.213344)"
  }
},
{
  "model": "tracking.event",
  "pk": 9801216,
  "fields": {
    "customer": 9,
    "fundraiser": 37096,
    "name": "conversation-end",
    "event_type": "click",
    "timestamp": "2026-03-04T09:40:12.665Z",
    "event_context": {
      "reason": "donation",
      "duration": 1.8100833333333335,
      "tag_name": null,
      "source_id": "603",
      "target_id": null,
      "source_category": "campaign-form"
    },
    "browser_location": {
      "current": "https://staging.formunauts.com/next/forms/603/form/pages/signature-step",
      "referrer": "https://staging.formunauts.com/next/forms/603/form/pages/qa-step"
    },
    "device_coordinates": "SRID=4326;POINT (16.365432 48.205567)"
```

```
  }
}
]
```